

Dynamic tech-for-good leader harnessing AI and data to drive global impact. Worked with the founders at Fjelltopp to transform the company from losses to profitability, spearheading public health data products used across 50+ countries. Founded gliff.ai going from concept to MedTech AI platform being used by 100s of NHS doctors and Fortune 500 companies. Published in good machine learning practices for pharmaceutical discovery (doi: <https://doi.org/10.1016/j.compbio.2024.108632>).

Experience

CEO / CTO (interim) · Data & AI Director · International Data Consultant

2022–2025

FJELLTOPP - SOCIAL ENTERPRISE & INTERNATIONAL DATA CONSULTANCY

- Developed and implemented new business strategies to increase revenue 200% to \$850k, turning losses into profit within 2 years.
- Managed \$1M of projects, with international stakeholders such as WHO, UNICEF, governments and healthcare companies.
- Provided fractional CTO leadership on public health data tools used by 50+ countries.
- Drove the adoption of international data standards across 14 government departments and NGOs for the Zambia National Research Repository.
- Provided data science consultancy of projects driving public health policy in Ukraine, Kazakhstan, Tanzania, Zambia and the WHO.

International Data Consultant

2023–2024

ECTAD BANGLADESH - PART OF THE UN'S FOOD & AGRICULTURAL ORGANISATION (FAO)

- Stabilised the Bangladesh Animal Health Informatics System, decreasing support calls from 30-40 per day to <1 per day.
- Reduced data failures from >10% of the 1,000 users per month to <1% within 3 months.
- Increased data completeness from 20% to >87% within 12 months.
- Coordinated and led a team of 8 national and international advisors implementing a new quality management system across the project.
- Collaborated with government agencies and partners to prioritise and deliver 10+ data visualisations for improved decision making.

Chief Technology Officer · Founder

2020–2022

GLIFF.AI - MEDTECH AI STARTUP

- Spun gliff.ai out of Durham university fundraising over £1M for a B2B SaaS MedTech AI product.
- Built the company into a high-growth SME with a 20x increase in valuation over 2 years.
- Pioneered development of new MedTech AI software impacting major healthcare project involving 100s NHS doctors.
- Invited to speak on AI regulation in healthcare at the 2022 Scottish AI Summit.
- Author on the Pistoia Alliance's Toolkit for Implementing Best Practices in AI and ML in drug discovery.
- Led our diversity strategy, working with our team of 15, leading to gliff.ai becoming a signatory of the Tech Talent Charter.

LKAS Research Fellow · EPSRC Doctoral Prize Research Fellow · Postdoctoral Research Associate

2017–2020

UNIVERSITY OF GLASGOW

- Awarded 2 fully-funded independent research fellowships focussed on data science for biomedicine.
- Developed bioimage analysis and graph theory approaches for understanding zebrafish brain responses in 3D images.
- Built collaborations across the world, including at the Max Planck Institute of Neurobiology - one of the top 20 research bodies globally.
- Invited to speak at 10+ local and national public science events including Pint of Science and the Institute of Physics Public Lecture Series.

Education

PhD in Computer Science · Master of Science in Biology and Physics within the Natural Sciences Programme

2009–2017

DURHAM UNIVERSITY

- PhD in bioimage informatics; Thesis title: *Mathematical morphology for quantification in biological and medical image analysis*.
- Developed and deployed computer vision and data science solutions with collaborators in academia and the NHS.
- Published 10 papers, 2 of which were in top 5 computer vision journals.
- First Class Honours MSci; Masters thesis title: *Scratching away at cancer with image analysis*.

Programming Python | Hugging Face · Jupyter · Ollama · Pandas · Plotly · PyTorch · SciPy · Seaborn · scikit-learn · spaCy · SQLAlchemy · Tensorflow

Technologies Artificial Intelligence | Classic machine learning · Deep learning · CNNs · LLMs · RAG · Knowledge Graphs · Natural Language Processing

Soft skills Data Analysis · Knowledge Transfer · Leadership · Multi-stakeholder engagement · Project management · Research · Strategy development